

First Course Handout

Course:

SPA611: Radio Astronomy

Even Semester, 3 - 0 - 0 -0 (9)

Schedule: M (11-12), ThF(10-11)

Venue: T204

Office Hours: Tuesdays 1800 - 1900 hrs at ESB2, 6th Floor, Room no. 717/715

Students can visit the instructor to discuss things related to the course. They are requested to drop an email by 12 pm on Tuesday if they intend to come.

Assignments are expected to be submitted within a week of the release of the questions. Late submission will incur a 1 mark penalty for each additional day from the next day.

Programming Assignments will be submitted within 10 days of the assignment, with a penalty of 1 mark per additional day. The codes have to be uploaded under a folder of your name in the GitHub repository of the course. Please contact the Instructor or TAs to be added to the repository as a contributor.

Quizzes will be conducted at the end of each module.

Objectives:

This course is an introductory course in radio astronomy. The course aims to expose the students to the nature of radio emission from natural objects, provide a detailed understanding of the design and development of novel instrumentation and techniques to study them.

Course Contents:

Module 1 (7 Lectures) Fundamentals of radio astronomy

Module 2 (6 Lectures) Understanding the properties of radio signals

Module 3 (12 Lectures) Antennas, receivers and single dish observations

Module 4 (7 Lectures) Fundamentals of Interferometry

Module 5 (6 Lectures) Calibration and Imaging with Interferometers

Grading Policy:

Quizzes at end of each module, Assignments and Programming Assignments: 30%

Mid Sem: 30 %

End Sem: 40 %

Ethics:

Cheating of any form in assignments or tests will not be tolerated. Nominal use of AI is allowed in assignments and programming assignments, but entire codes/home-work done by AI will incur 0 marks for the entire assignment.

Reading material:

Interferometry and Synthesis in Radio Astronomy, Thomson, Moran and Swenson,
<https://link.springer.com/book/10.1007/978-3-319-44431-4>

Essential Radio Astronomy, Condon and Ranson,
<https://link.springer.com/book/10.1007/978-3-540-85122-6>

Tools of Radio Astronomy, Wilson, Rohlfs and Huttemeister,
<https://link.springer.com/book/10.1007/978-3-319-44431-4>